

Unit 4 Guided Notes — Geometry: Angles & Triangles

Name the relationship, write the equation, solve for x . Angles always add up to something.

 **By the end of this unit, I can...**

- Use angle relationships formed by parallel lines and a transversal
- Apply the Triangle Angle-Sum and Exterior-Angle Theorems
- Solve for unknown angles by writing and solving equations

Standard: NC.8.G.5



Blank Worksheet (PDF)



Answer Key (PDF)



Official PDF Packet

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in. Name the relationship, write the equation, solve for x — angles always add up to something!

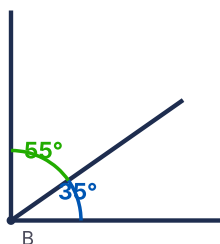
1 Angle Relationships

Fill the definitions and rules on your printed copy, then click to check.

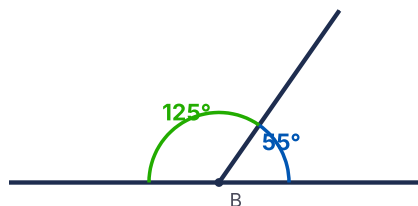
Relationship	Definition	Sum / Rule
Complementary	Two angles that add to °	
Supplementary	Two angles that add to °	
Vertical Angles		
Linear Pair		

Each angle is named with three letters — the vertex is the MIDDLE letter (e.g. $\angle ABD$ has vertex B). Arcs show which angle each measure belongs to.

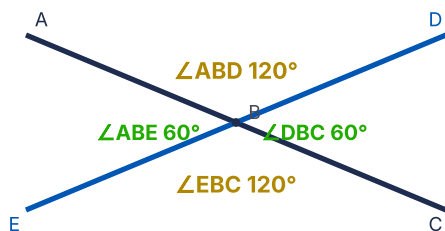
COMPLEMENTARY = 90°



SUPPLEMENTARY = 180°

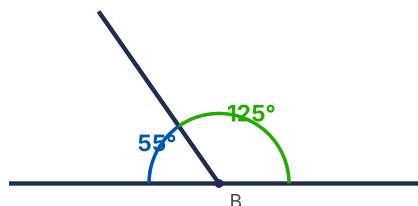


VERTICAL ANGLES



Opposite angles are EQUAL: $\angle ABD = \angle EBC$, $\angle DBC = \angle ABE$

LINEAR PAIR = 180°



Quick Check

MC Two angles form a linear pair. One measures 115° . What is the other?

✓ A linear pair adds to 180° . $180^\circ - 115^\circ = 65^\circ$.

Match the Relationship to its Rule

MATCH Click a relationship, then its rule.

Relationship

Complementary

Supplementary

Vertical Angles

Linear Pair

Rule

Always EQUAL

Add to 90°

Adjacent angles on a line $\rightarrow 180^\circ$

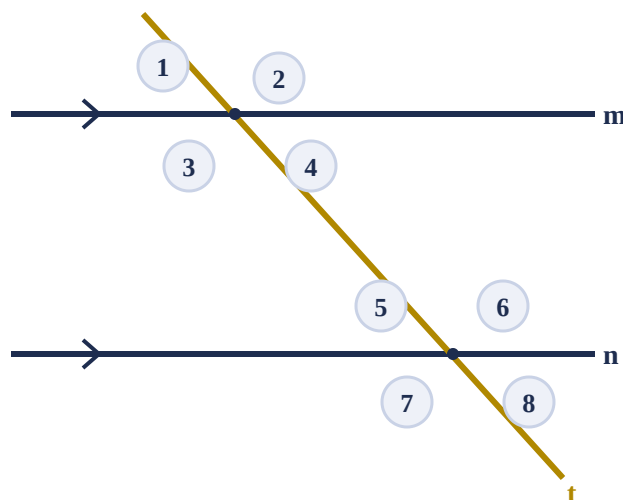
Add to 180°

0 of 4 matched

2 Parallel Lines & Transversals

REMEMBER: Corresponding, Alternate Interior, Alternate Exterior, and Vertical angles are CONGRUENT (equal). Same-Side Interior and Same-Side Exterior angles are SUPPLEMENTARY (add to 180°).

PARALLEL LINES & TRANSVERSAL



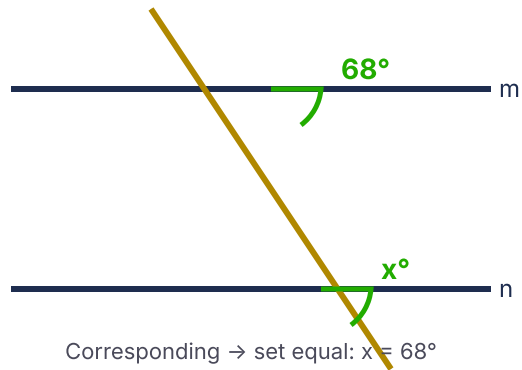
Two parallel lines ($m \parallel n$) cut by transversal t form 8 angles. Corresponding: (1,5)(2,6)(3,7)(4,8) · Alt. Interior: (3,6)(4,5) · Alt. Exterior: (1,8)(2,7) · Same-Side Interior: (3,5)(4,6)

Fill the Relationship and Rule columns on your printed copy, then click to check.

Angle Pair Type	Relationship	Rule to Use	Example
Corresponding			$68^\circ \rightarrow x = 68^\circ$
Alternate Interior			$101^\circ \rightarrow x = 101^\circ$
Alternate Exterior			$142^\circ \rightarrow x = 142^\circ$
Vertical			$120^\circ \rightarrow x = 120^\circ$

Same-Side Interior	<u> </u>	<u> </u>	$134^\circ \rightarrow x = 46^\circ$
Same-Side Exterior	<u> </u>	<u> </u>	$77^\circ \rightarrow x = 103^\circ$

CORRESPONDING ANGLES ARE EQUAL



Worked Example — $\angle 1 = 3x + 15$ and $\angle 2 = 5x - 7$ (alternate interior angles)

Alternate interior angles are EQUAL: $3x + 15 = 5x - 7$

Subtract $3x$: $15 = 2x - 7$

Add 7: $22 = 2x$

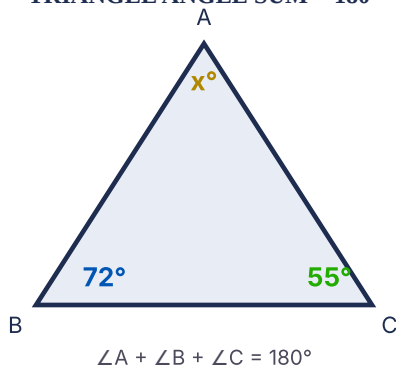
✓ **Divide by 2:** $x = 11 \rightarrow \text{angle} = 3(11) + 15 = 48^\circ$

3 Triangle Theorems

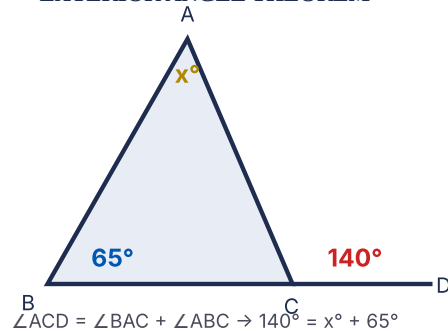
TRIANGLE ANGLE SUM: The three interior angles of a triangle always add up to 180° .

EXTERIOR ANGLE THEOREM: An exterior angle equals the SUM of the two non-adjacent (remote) interior angles.

TRIANGLE ANGLE SUM = 180°



EXTERIOR ANGLE THEOREM



Problem 1 — Triangle Angle Sum: $x + 72 + 55 = 180$

Solve: $x + 127 = 180 \rightarrow x = 53^\circ$

Problem 2 — Exterior Angle Theorem: $x + 65 = 140$

Solve: $x + 65 = 140 \rightarrow x = 75^\circ$

ERROR ANALYSIS Jordan's Exterior Angle Mistake — A triangle has interior angles 35° and 70° . Exterior angle = x . Jordan wrote

$$x = 70 - 35 = 35^\circ$$

Jordan subtracted instead of adding. The exterior angle = SUM of the remote interior angles: $x = 35 + 70 = 105^\circ$.



Go to Unit 4 Practice →



Part 2 — Practice Problems

 **Identify the relationship, write the equation, and solve for x .** Type just the number, e.g. 55.

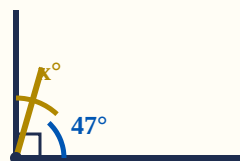
A. Angle Relationships — Solve for x

1) Supplementary Angles, given 125° . $x = ?$



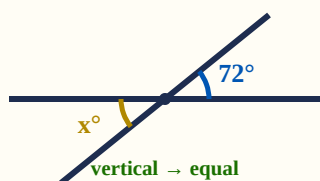
$x =$

2) Complementary Angles, given 47° . $x = ?$



$x =$

3) Vertical Angles, given 72° . $x = ?$



$x =$

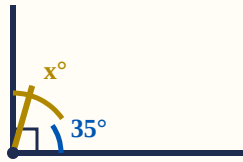
4) Linear Pair, given 138° . $x = ?$



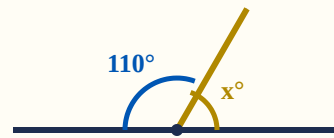
$x =$

5) Complementary Angles, given 35° . $x = ?$

6) Supplementary Angles, given 110° . $x = ?$

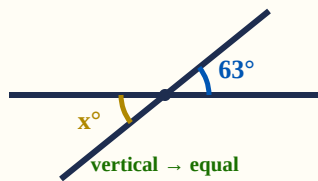


$x =$



$x =$

7) Vertical Angles, given 63° . $x = ?$



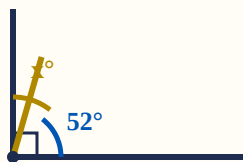
$x =$

8) Linear Pair, given 157° . $x = ?$



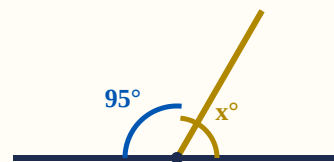
$x =$

9) Complementary Angles, given 52° . $x = ?$



$x =$

10) Supplementary Angles, given 95° . $x = ?$

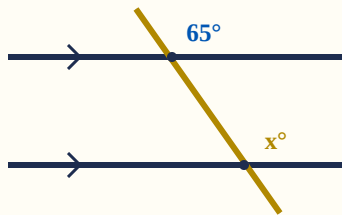


$x =$

B. Parallel Lines & Transversals

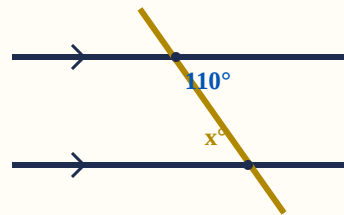
1) Corresponding, given 65° . $x = ?$

2) Alternate Interior, given 110° . $x = ?$



Corresponding $\rightarrow$ equal

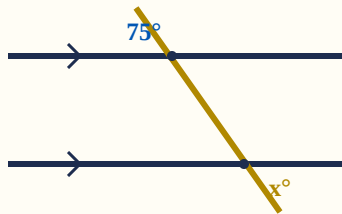
$x =$



Alternate Interior $\rightarrow$ equal

$x =$

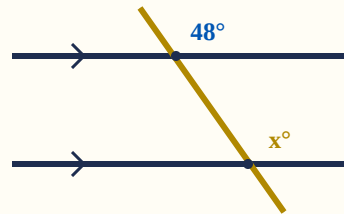
3) Alternate Exterior, given 75° . $x = ?$



Alternate Exterior $\rightarrow$ equal

$x =$

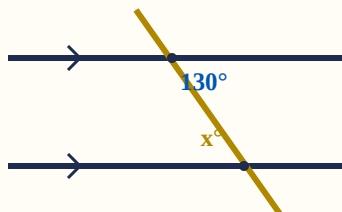
4) Corresponding, given 48° . $x = ?$



Corresponding $\rightarrow$ equal

$x =$

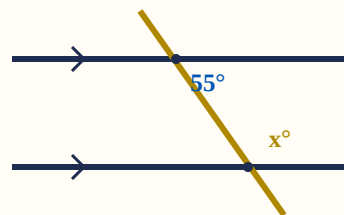
5) Alternate Interior, given 130° . $x = ?$



Alternate Interior $\rightarrow$ equal

$x =$

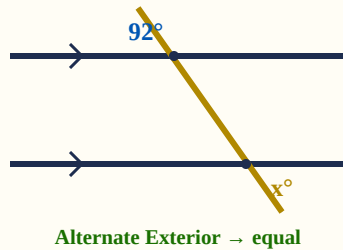
6) Co-Interior, given 55° . $x = ?$



Co-Interior $\rightarrow$ add to 180°

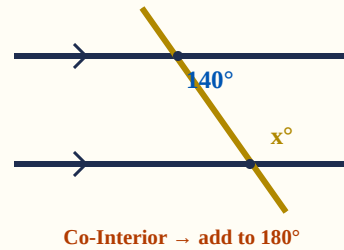
$x =$

7) Alternate Exterior, given 92° . $x = ?$



$x =$

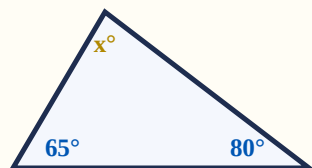
8) Co-Interior, given 140° . $x = ?$



$x =$

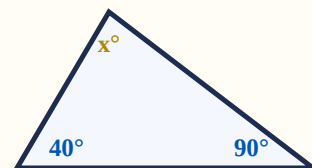
C. Triangle Theorems

1) Angle Sum, given 65° , 80° . $x = ?$



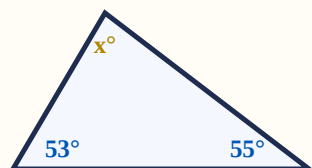
$x =$

2) Angle Sum, given 40° , 90° . $x = ?$



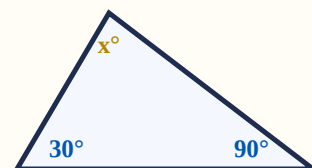
$x =$

3) Angle Sum, given 53° , 55° . $x = ?$



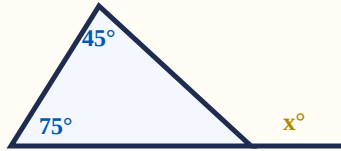
$x =$

4) Angle Sum, given 30° , 90° . $x = ?$



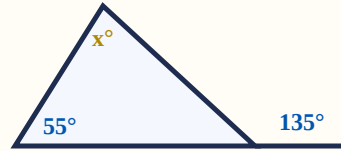
$x =$

5) Exterior Angle Theorem, remotes
 45° , 75° . Exterior $x = ?$



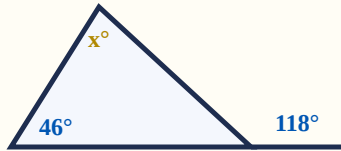
$x =$

6) Exterior Angle Theorem, remote
 55° , exterior = 135° . $x = ?$



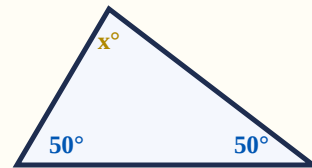
$x =$

7) Exterior Angle Theorem, remote
 46° , exterior = 118° . $x = ?$



$x =$

8) Angle Sum, given 50° , 50° . $x = ?$



$x =$

[← Back to Unit 4](#)